

a. Find $\frac{dy}{dx}$ if $y = 2^{\cos x + \ln x}$.

[7 Marks]

Solution:

Method I: Logarithmic Differentiation

$$\ln y = \ln 2^{\cos x + \ln x} = (\cos x + \ln x) \ln 2.$$

Implies

$$\frac{1}{y} \frac{dy}{dx} = \frac{d}{dx} [(\cos x + \ln x) \ln 2] = \ln 2 \left[\frac{d}{dx} \cos x + \frac{d}{dx} \ln x \right] = \ln 2 \left(-\sin x + \frac{1}{x} \right).$$

Implies

$$\frac{dy}{dx} = y \left[\ln 2 \left(-\sin x + \frac{1}{x} \right) \right] = 2^{\cos x + \ln x} \left[\ln 2 \left(\frac{1}{x} - \sin x \right) \right].$$

$$\frac{dy}{dx} = (\ln 2) 2^{\cos x + \ln x} \left(\frac{1 - x \sin x}{x} \right).$$

Method II: In Exponential Form [$a^x = e^{x \ln a}$]

$$y = e^{(\cos x + \ln x) \ln 2}.$$

So,

$$\frac{dy}{dx} = e^{(\cos x + \ln x) \ln 2} \frac{d}{dx} [(\cos x + \ln x) \ln 2] = e^{(\cos x + \ln x) \ln 2} \left[\ln 2 \left(-\sin x + \frac{1}{x} \right) \right].$$

$$\frac{dy}{dx} = 2^{\cos x + \ln x} \left[\ln 2 \left(-\sin x + \frac{1}{x} \right) \right].$$

b. Find $\lim_{x \rightarrow 0} (1 + \sin x)^{1/x}$.

[6 Marks]

Solution. As discussed above, we begin by introducing a dependent variable

$$y = (1 + \sin x)^{1/x}$$

and taking the natural logarithm of both sides:

$$\ln y = \ln(1 + \sin x)^{1/x} = \frac{1}{x} \ln(1 + \sin x) = \frac{\ln(1 + \sin x)}{x}$$

Thus,

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x}$$

which is an indeterminate form of type $0/0$, so by L'Hôpital's rule

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x} = \lim_{x \rightarrow 0} \frac{(\cos x)/(1 + \sin x)}{1} = 1$$

Since we have shown that $\ln y \rightarrow 1$ as $x \rightarrow 0$, the continuity of the exponential function implies that $e^{\ln y} \rightarrow e^1$ as $x \rightarrow 0$, and this implies that $y \rightarrow e$ as $x \rightarrow 0$. Thus,

$$\lim_{x \rightarrow 0} (1 + \sin x)^{1/x} = e \quad \blacktriangleleft$$